Engineering Solution Option Appraisal

Mapping: K1, K2, A1 Suggested time: 35-60 minutes

Executive brief

Compare closed-loop water, heat recovery and material substitution options. Endorse one option subject to evidence gates.

Deliverable

Prepare a one-page decision record containing the recommendation, calculation, evidence sources, assumptions, control owner, approval condition and residual uncertainty.

Procedure

1. Inspect mock-data/mock-data.csv; confirm units, time period and boundary.
2. Identify missing, estimated or stale evidence.
3. Apply the calculation or prioritisation rule described in the Learner Guide.
4. Test at least one alternative assumption.
5. Record the decision and the conditions that must be satisfied before external reporting.
6. Peer-review the work: another group must be able to reproduce the result.

Acceptance criteria

- Every claim traces to a named mock-data row or cited source.
- Units and formulas reconcile.
- Thresholds and assumptions are explicit.
- The recommendation states commercial, ecological and community implications.
- No personal data or completed learner record is used.

References

- Official reference: https://www.iea.org/reports/energy-efficiency-2023
- Course trainer slides v5.0
- Learner Guide v5.0

Detailed worksheet

Use WORKSHEET.md for the activity-specific formulas, thresholds, worked example and results table. Use references/SOURCES.md for the reference brief.